1 Fig. 1.1 shows a thin cylindrical metal rod of length *L*.

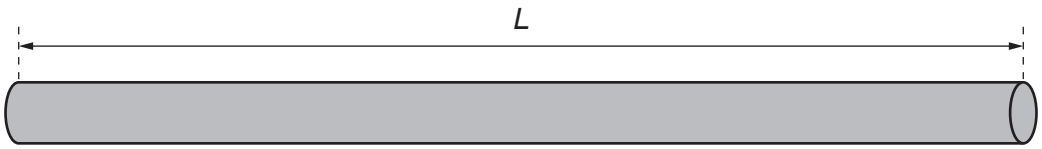


Fig. 1.1

One end of the rod is hit with a hammer. A stationary sound wave is set up within the rod. The rod vibrates at its resonant frequency *f*.

A microphone placed at the other end of the rod detects the sound wave emitted from the rod. The frequency of the detected sound is also *f*.

A number of rods of different length are available.

It is suggested that *f* is related to *L* by the relationship

$$2fL^n = \sqrt{\frac{E}{\rho}}$$

where ρ is the density of the metal, and *E* and *n* are constants.

Plan a laboratory experiment to test the relationship between *f* and *L*.

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for *E* and *n*.

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

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